

Visualization of Complex Data

DATS6401

Final Term Project (FTP)

The objective of FTP is to apply the Python visualization techniques covered in the course to a real-world dataset.

Develop a Python web-based dashboard (app) for interactive information visualization and storytelling. Deploy the developed app through Google Cloud Platform (GCP) for production. The final term project has three phases:

Graph & Figure Formatting Requirements (all phases): All graphs must include a title. Numbers must be large enough to be readable. Display all floating-point numbers with 2-digit decimal precision. **Title** = [font: your choice | color: your choice | size: large enough]. **X, Y label** = [font: your choice | color: your choice | size: large enough]. **Font family** = [your choice]. **Color** = [your choice]. **Theme** = [your choice].

Phase I: Static graphs & Tables[35pts]. You need to explore the dataset and reveal underlying information by plotting static plots for the following feature types. You may need to clean the dataset, if necessary, in this phase.

- Categorical features [for bar plot it must be sorted and numbers displayed on the graph]
- Numerical features

You are free to use any package that you feel comfortable with for this phase. The plots in this phase are static which means a plot cannot be updated without re-execution of the code. The following list of plots needs to be included in this phase for exploring the selected dataset.

- Line-plot
- Bar plot: stack, group
- Count plot
- Pie chart
- Dist plot
- Pair plot
- Heatmap with cbar
- Histogram plot with KDE
- QQ-plot
- KDE plot with fill, chose an alpha, choose a palette, choose a linewidth
- Im or reg plot with scatter representation and regression line
- Multivariate Box plot
- Multivariate Boxen plot
- Area plot
- Violin plot
- Joint plot with KDE and scatter representation
- Rug plot
- 3D plot and contour plot
- Cluster map
- Hexbin

- Strip plot
- Swarm plot

Plotting a static graph & tables for this phase is necessary but not sufficient. The **sufficient condition is the explanation of each plot and table** (pretty table or tabulate). A plot or table without explanation or inferences will receive zero credit. In this phase you also need to include subplots (at least 4) for storytelling i.e., multiple pie/bar plots in one subplot that tell a story about features inside the dataset, then write your observations. You need to include legend (hue) in your static plot for comparison. All figures must include title, legend, x-label, y-label, and grid and they must be customized with assorted color, font size, line width, ... (see the Formatting Requirements above). Make sure the information on the graph is not blocked.

Phase II: Interactive web-based dashboard[35pts]. In this phase you are required to use Dash framework with **dash-bootstrap-components (dbc)** for an interactive web-based dashboard. **All components must be interactive**, and **each question must be placed under a separate Tab using dbc.Tabs**. All the plots in the previous phase need to become interactive through a dashboard in this phase. In an interactive dashboard, a user can update a plot without re-executing the code. A user must be able to switch the input, and the graph should be updated accordingly. There is no need to re-execute the Python code to update a plot in this phase. All **float numbers must be displayed with 2-digit decimal places**. You are required to use Bootstrap components from dash-bootstrap-components. For reference, see <https://dash-bootstrap-components.opensource.faculty.ai/>. The bare minimum **dbc** and **dcc** components are.

- dbc.Container – main layout wrapper for the entire application
- dbc.Card, dbc.CardBody, dbc.CardHeader – grouping and presenting each section or question
- dbc.Tabs, dbc.Tab – each question must be placed in a separate Tab
- dbc.Select – dropdown selection for features, plot types, and methods
- dbc.RadioItems – radio button selection between methods or options
- dbc.DropdownMenu, dbc.DropdownMenuItem – menu-based feature or method selection
- dbc.Button – trigger callbacks and user interactions
- dbc.Checklist – multi-option selection (e.g., selecting multiple features)
- dbc.Tooltip – informational tooltips displayed on hover
- dbc.Badge – display computed numerical values formatted to 2-digit decimal places
- dbc.Alert – display status messages, warnings, or results
- dbc.Row, dbc.Col – responsive grid layout for organizing dashboard content
- dbc.Spinner – loading indicator for long-running computations
- dcc.Graph – interactive Plotly graphs
- dcc.Slider, dcc.RangeSlider – range and value selection
- dcc.Download – file download functionality
- dcc.Textarea – text input and output areas
- html.H1–H6, html.Div, html.Label, html.Img – HTML structural and layout elements
- dbc.NavBar, dbc.NavItem – navigation bar for the application
- Dash DAQ (dash_daq) – a suite of robust interactive industrial controls. Required components include: daq.Knob (rotary control), daq.Gauge (circular gauge display)

- u. `daq.Thermometer` (thermometer-style indicator),
- v. `daq.LEDDisplay` (LED-style numeric display – use for showing 2-decimal float values),
- w. `daq.BooleanSwitch` (on/off toggle),
- x. `daq.ColorPicker` (interactive color selector),
- y. `daq.GraduatedBar` (graduated bar indicator).

See <https://dash.plotly.com/dash-daq>

You need to design an interactive Python-based front end (dashboard) that interactively plots the data. You need to start with the draft of the dashboard submitted in the proposal. The dashboard must be structured using `dbc.Container` as the outermost wrapper, and each of the following questions must be placed under a separate `dbc.Tab` inside `dbc.Tabs`. Each `Tab` must contain at least one `dbc.Card` wrapping a `dbc.CardHeader` and `dbc.CardBody`. All float numbers displayed in the dashboard must be formatted to 2-digit decimal places. The required `Tabs` and their content are: Load Data, Data Cleaning [different methods selectable via `dbc.RadioItems` or `dbc.Select`], Outlier Detection & Removal [different methods selectable], Dimensionality Reduction (PCA), Normality Tests [different methods selectable], Data Transformation [different methods selectable], Numerical Features Plots [user selectable feature(s) and plot type via `dbc.Select` or `dbc.DropdownMenu`], Categorical Features Plots [user selectable feature(s) and plot type], and Statistics.

Phase III: Deployment [10pts]: In this phase you need to dockerize the developed app and deploy it through Google Cloud Platform for production. The GCP needs to be used for deployment. [if you want to use different cloud platforms, you need to get pre-approval]. You need to include the working worldwide web address (www) of your term project at the top of the report.

If you need more clarification, please reach out to me. The required software for FTP is Python only and the required IDE is PyCharm [only]. You can use Tableau in parallel for the verification of the results, but all graphs must be generated in Python.

A formal report [latex or word in the .pdf format], presentation and demo of the created app using Google Cloud Platform (GCP) is required by the deadline. API format for citations.

SPECIFIES

The final formal report must be typed and should contain the following sections. You are free to place phase I, II and III requirements under an appropriate section in the report as shown below.

- 1- **Cover page.** Name of the school, author, instructor, date, course name, and number.
- 2- **Table of content.**
- 3- **Table of figures and tables.**
- 4- **Abstract.**
- 5- **Introduction.** An overview of the procedures to accomplish the FTP objectives and an outline of the report.
- 6- **Description of the dataset [5pts]:** You need to provide a description of the selected dataset and how the dataset satisfies the dataset criteria. You need to specify which variable in the selected dataset will serve as dependent variable and which ones serve as independent variables. You will need to explain the importance of the selected dataset in industry. (both static and interactive)

- 7- **Pre-processing dataset [5pts]:** Data cleaning for missing samples, NAN's. Explain which method was used for data cleaning. You will need to display the first few observations of the cleaned dataset and the corresponding statistics. (both static and interactive)
- 8- **Outlier detection & removal [5pts]:** Use one of the methods explained in class for the outlier detection and removal from the raw dataset. Write down your observations. (both static and interactive)
- 9- **Principal Component Analysis (PCA) [5pts]:** Perform a complete PCA analysis of the cleaned dataset for a feature dimension reduction. Include the complete explanation in your report. Check the condition number and the singular values of the reduced dimension features. You can replace PCA with a random forest method. Write down your observations. (both static and interactive)
- 10- **Normality test [5pts]:** Use one of the tests explained in class to see if the dataset comes from the Gaussian distribution or not. Write down your observations. (both static and interactive)
- 11- **Data transformation:** If transformation of the dataset is needed, you need to explain which method was used. For example: non-gaussian to gaussian distribution transformation. (both static and interactive)
- 12- **Heatmap & Pearson correlation coefficient matrix [5pts]:** Display the Pearson correlation coefficient between variables using heatmap and scatter plot matrix. Explain your observations through a table. (both static and interactive)
- 13- **Statistics [5pts]:** You need to statistically analyze the dataset and write down your observations accordingly. Use the statistics tools discussed in class. You will need to display the estimated multivariate kernel density estimate. (both static and interactive)
- 14- **Data visualization:** Visualize the dataset using the static plots in phase II and discuss what can be observed from each plot. You need to write down your observations for each plot. (both static and interactive).
- 15- **Subplots:** (both static and interactive).
- 16- **Tables:** (both static and interactive)
- 17- **Dashboard [35pts]:** Take a screenshot of the designed dashboard [each tab as a separate screenshot] and place it into your report. The functionality of the designed dashboard needs to be demonstrated during the presentation. The dashboard must contain all sections of the static phase but in the interactive format of the dashboard.
- 18- **Observations:** For every graph in your report, you need to provide your observations. Plots without observations will receive zero points. The main objective is to use Python to understand datasets, and you must reveal underlying information using visualization tools.
- 19- **The conclusion** is an important section of your final report which could include the following:
 - a. What did you learn from various graphs created in this project?
 - b. How does the created Python dashboard help users gain information from the selected dataset?
 - c. Is the created dashboard user-friendly? You can share the created app on professional social media [e.g., LinkedIn] and ask people for comments, then include their feedback in your report. Make sure to get permission from the commenters before publishing their responses.
 - d. Functionality: How functional is the created dashboard?
- 20- A **separate appendix** should contain the supporting Python code developed for this project.
- 21- **References**

Submission Notes

- The **soft copy of your Python programs** needs to be submitted separately as a .py to verify the results in the report. Make sure to include the dataset in your submission. Make sure to run your code before submission. If the Python code generates an error message, 50% of the term project points will be forfeited.
- Include a **readme.txt** file that explains how to run your Python code. All results in your report must be reproducible to receive full credit.
- The FTP is an individual assignment. All coding must be done individually and must be original. You are not allowed to use any AI-based resources, i.e., ChatGPT, for the term project. Copying code from the internet without proper citation will be considered **plagiarism** and FTP grade will be disregarded. Make sure to write your own code to avoid future complications.
- All figures in your report must include a proper x-label, y-label, title, and legend [if applicable]. Pick an appropriate theme or style for the plotted graphs. Provide a proper title for each table included in your report.
- **Final presentation & Demo:** The presentation weighs 20% of the term project grade. You have 20 minutes to present your term project and demonstrate the functionality of the developed dashboard to the class. You need to record your presentation [need a PowerPoint] at home and submit the recorded presentation and the PowerPoint through BB by the deadline. The due date by **Tuesday, May 5th** .
- **Final report** [the PDF and the Python file] is due by **Wednesday, May 6th** .
- Upload the **final report (as a single pdf)** plus **the .py file(s)** through BB by the due date.